

Model Validation, Overfitting Control & Hyperparameter Tuning

Artificial Intelligence & Machine Learning – Task 3 Report

1. Introduction

Machine Learning models often perform well on training data but may fail to generalize to unseen data. This phenomenon is known as overfitting and is one of the most common challenges in real-world machine learning projects. To build reliable and production-ready models, it is important to validate model performance properly and apply techniques that improve generalization.

In this project, model validation, overfitting detection, cross-validation, and hyperparameter tuning techniques were implemented using the California Housing Dataset. The project extends the work completed in Task 2 by focusing on model reliability rather than only prediction accuracy.

The main objective of this project is to evaluate model performance scientifically, reduce overfitting, and select the most reliable regression model through systematic validation and optimization.

2. Objectives

The primary objectives of this project are:

- To understand overfitting and underfitting in machine learning models.
 - To evaluate model performance using cross-validation.
 - To compare training and testing errors for overfitting detection.
 - To optimize model performance using GridSearchCV.
 - To tune Decision Tree hyperparameters systematically.
 - To compare baseline and optimized models.
 - To justify final model selection using objective evaluation metrics.
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3. Dataset Description

The California Housing Dataset available in the scikit-learn library was used for this project. The dataset contains housing-related information collected from districts in California and is widely used for regression analysis.

Features Included

Feature	Description
MedInc	Median income
HouseAge	Average house age
AveRooms	Average number of rooms
AveBedrms	Average number of bedrooms
Population	Total population
AveOccup	Average occupancy
Latitude	Latitude coordinate
Longitude	Longitude coordinate

Target Variable

HousePrice (Median House Value)

The goal of the model is to predict house prices using the available housing characteristics.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed before model training to understand the structure and quality of the dataset.

The following analyses were conducted:

- Dataset shape analysis
- Feature inspection
- Statistical summary
- Missing value analysis
- Correlation analysis
- Visualization of relationships among features

Observations

- The dataset consists entirely of numerical variables.
- No significant missing values were observed.
- Median Income showed a strong positive correlation with house prices.

- Geographic variables such as Latitude and Longitude significantly influenced house values.
- Population had a weaker relationship with housing prices.

A correlation heatmap was generated to visualize relationships between variables.

5. Data Preprocessing

Proper preprocessing is essential for building stable machine learning models.

The following preprocessing steps were performed:

Feature Separation

The target variable was separated from the independent variables.

Feature Scaling

StandardScaler was applied to normalize all feature values.

Benefits of feature scaling include:

- Improved model stability
- Better convergence during training
- Fair comparison among features
- Reduced influence of feature magnitude differences

Train-Test Split

The dataset was divided into:

- 80% Training Data
- 20% Testing Data

This ensured unbiased model evaluation on unseen data.

6. Baseline Model Development

Three regression models from previous tasks were used as baseline models:

Linear Regression

A simple regression model that assumes a linear relationship between features and target values.

Ridge Regression

An extension of Linear Regression that applies regularization to reduce overfitting.

Decision Tree Regressor

A tree-based model capable of capturing complex nonlinear relationships.

These models provided the foundation for further validation and optimization.

7. Overfitting Detection

Overfitting occurs when a model performs extremely well on training data but poorly on unseen testing data.

To detect overfitting, training RMSE and testing RMSE were compared using the Decision Tree model.

Analysis Method

- Train the Decision Tree model.
- Calculate Training RMSE.
- Calculate Testing RMSE.
- Compare both values.

Interpretation

- Small difference between train and test RMSE indicates good generalization.
- Large difference indicates overfitting.
- Excessively complex tree structures are more likely to overfit.

A Train RMSE vs Test RMSE comparison chart was generated to visualize overfitting behavior.

8. Cross-Validation

A single train-test split may not provide reliable performance estimates. Therefore, cross-validation was applied.

5-Fold Cross Validation

The dataset was divided into five folds:

1. Four folds used for training.
2. One fold used for validation.
3. Process repeated five times.
4. Average performance calculated.

Benefits of Cross Validation

- Reduces dependency on a single train-test split.
- Provides more stable performance estimates.
- Improves confidence in model reliability.
- Commonly used in industry-level machine learning projects.

Cross-validation RMSE was calculated to evaluate model consistency.

9. Hyperparameter Tuning Using GridSearchCV

Hyperparameters control model behavior and significantly influence performance.

GridSearchCV was used to identify the optimal Decision Tree configuration.

Parameters Tuned

max_depth

Controls maximum tree depth.

Values tested:

- 3
- 5
- 7

- 10

min_samples_split

Controls minimum samples required for node splitting.

Values tested:

- 2
- 5
- 10

GridSearchCV Process

- Multiple parameter combinations were evaluated.
- Cross-validation was applied internally.
- Best parameter combination was selected automatically.

This systematic search improved model performance and reduced overfitting.

10. Tuned Model Evaluation

After hyperparameter tuning, the optimized Decision Tree model was evaluated using:

Root Mean Squared Error (RMSE)

RMSE measures average prediction error.

Lower RMSE indicates better performance.

R² Score

R² Score measures the proportion of variance explained by the model.

Higher R² values indicate stronger predictive capability.

The tuned model was then compared with the baseline models developed in previous tasks.

11. Model Comparison and Final Selection

The following models were compared:

- Linear Regression
- Ridge Regression
- Tuned Decision Tree Regressor

Comparison Metrics

- RMSE
- R^2 Score
- Cross Validation RMSE
- Overfitting Analysis

Selection Criteria

The final model was selected based on:

- Lowest prediction error
- Highest explanatory power
- Stable cross-validation performance
- Reduced overfitting

The model comparison table and visualization charts clearly demonstrated the strengths and weaknesses of each model.

12. Results and Discussion

The tuned Decision Tree model demonstrated improved performance compared to the untuned version.

Cross-validation provided a more realistic estimate of model performance than a single train-test split.

Hyperparameter tuning helped:

- Improve generalization
- Reduce overfitting
- Increase model reliability

Residual analysis further confirmed that prediction errors were distributed more consistently around zero, indicating a well-performing model.

13. Improvement Ideas

The following techniques can further improve performance:

- Random Forest Regression
- Gradient Boosting Regression
- XGBoost Regression
- Advanced Feature Engineering
- Feature Selection Techniques
- RandomizedSearchCV
- Nested Cross Validation
- Ensemble Learning Methods

These approaches may provide higher predictive accuracy and robustness.

14. Conclusion

In this project, model validation, overfitting control, and hyperparameter tuning techniques were successfully implemented using the California Housing Dataset.

Overfitting was analyzed by comparing training and testing performance. Cross-validation was used to obtain reliable performance estimates, while GridSearchCV was applied to optimize Decision Tree hyperparameters systematically.

The optimized model was compared with baseline models using RMSE and R^2 Score, allowing an objective and scientific model selection process.

This project provided practical experience in professional machine learning workflows and strengthened understanding of model validation, generalization, overfitting control, and hyperparameter optimization. It also highlighted the importance of selecting models based on reliability and stability rather than relying solely on accuracy metrics.